

Test Queries Report

Live Data AI Assistant — Telegram Bot (n8n + Apify + Gemini)

This report documents five end-to-end test queries executed against the deployed Telegram bot, covering both the LIVE data path (routed through Apify and summarized by Gemini) and the GENERAL knowledge path (answered directly by Gemini). Each test shows the exact input sent, the routing decision made by the classifier, the category assigned, the full response delivered back to the user on Telegram, and the logging status recorded in Google Sheets.

#	Input Query	Expected Route	Actual Route	Category	Status
1	latest data center news today	LIVE	LIVE	NEWS	SUCCESS
2	current iPhone price	LIVE	LIVE	PRICE	SUCCESS
3	top remote jobs today	LIVE	LIVE	JOBS	SUCCESS
4	what is deep learning	GENERAL	GENERAL	NONE	SUCCESS
5	explain the difference between TCP and UDP	GENERAL	GENERAL	NONE	SUCCESS

All five queries were classified correctly by the AI router with no keyword-matching involved — routing and category assignment were both determined entirely by the Gemini classification call.

Detailed Input / Output Pairs

Test 1

Route	LIVE	Category	NEWS	Status	SUCCESS
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Input (sent via Telegram):

latest data center news today

Output (bot reply):

Hi there! Here is the latest on what's happening in the data center world today:

The industry is currently focused heavily on the intersection of AI, power infrastructure, and expansion challenges. Here are the key highlights:

- Grid and Power Pressures: Regulators (like NERC) are raising concerns about the strain AI data centers are putting on power grids, with Texas exploring new rules for co-locating campuses near power plants.
- Sustainability Scrutiny: Reports indicate that AI data centers are consuming significantly more water than previously estimated by many tech companies.
- Project Shifts: Major developments continue to fluctuate, such as a 29MW data center project in Montana being canceled after the building owner withdrew support.
- Global Growth: On the infrastructure side, companies like PRASA are expanding manufacturing capabilities, recently launching one of India's largest dedicated modular data center plants.

For more deep dives, you can check out Data Center Knowledge and Data Center Dynamics.

Test 2

Route	LIVE	Category	PRICE	Status	SUCCESS
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Input (sent via Telegram):

current iPhone price

Output (bot reply):

Hey there! Looking for current iPhone prices? Here's a quick summary from what I found:

- For the latest models, you can often find them with flexible monthly payment plans. For example, an iPhone 17 Pro Max 256GB was mentioned at around \$33.34 per month over 36 months with 0% APR at Best Buy.
- Apple offers Trade In programs where you can get credit towards a new iPhone (like iPhone 17, iPhone Air, or iPhone 17 Pro) by trading in an eligible older model (iPhone 13 or newer).
- If you're looking for more budget-friendly options, older or refurbished iPhones like the iPhone 12 series, iPhone SE 3, or iPhone SE 2 are also available at varying price points from other retailers.

Hope this helps you get started! You can check out Apple's official store or retailers like Best Buy for more details.

Test 3

Route	LIVE	Category	JOBS	Status	SUCCESS
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Input (sent via Telegram):

top remote jobs today

Output (bot reply):

Hey there! Here's a quick rundown of some top remote job opportunities you might find interesting today:

Trending Roles: AI Prompt Engineering, Freelance Writing, and Marketing Automation are seeing recent activity.

Hiring Companies: Keep an eye on big names like U.S. Bank, Salesforce, and Wikimedia Foundation, which are actively hiring remote talent.

Job Categories: Developer, Engineer, and Management/Operations roles are trending on platforms like Dynamite Jobs.

You can check out sites like We Work Remotely and Remote.co for more listings!

Test 4

Route	GENERAL	Category	NONE	Status	SUCCESS
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Input (sent via Telegram):

what is deep learning

Output (bot reply):

Hey there!

So, "deep learning" is a really cool type of machine learning that's all about teaching computers to learn from data in a way that's inspired by how the human brain works. Think of it like building an artificial "brain" with many layers (hence "deep").

Here's a quick rundown of what makes it special:

- Inspired by the Brain: It uses artificial neural networks, which have interconnected "neurons" that process information.
- Learns Hierarchies: Each layer in the network learns to recognize different levels of features. So, in an image, the first layers might detect edges, the next shapes, and so on, until it can recognize complex objects.
- Learns from Data: The more data you feed it, the better it gets at making predictions or decisions.
- Automatic Feature Extraction: Unlike traditional machine learning, you don't have to tell it exactly what features to look for. It figures them out on its own!

It's what powers a lot of the amazing tech we see today, like:

- Image and speech recognition (think voice assistants!)
- Natural language processing (like how chatbots understand you)
- Self-driving cars
- Recommender systems (like on Netflix or Spotify)

Hope that makes sense! Let me know if you have any more questions.

Test 5

Route	GENERAL	Category	NONE	Status	SUCCESS
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Input (sent via Telegram):

explain the difference between TCP and UDP

Output (bot reply):

Hi there! Think of the difference between TCP and UDP as the difference between sending a registered letter versus shouting across a room. Here is the breakdown:

TCP (Transmission Control Protocol)

TCP is all about reliability. It ensures every packet arrives in the correct order. If something goes missing, TCP requests a retransmission.

- **Reliable:** Guarantees delivery.
- **Ordered:** Reassembles data in the right sequence.
- **Connection-oriented:** Establishes a "handshake" before sending data.
- **Use cases:** Web browsing (HTTP/HTTPS), email (SMTP), and file transfers.

UDP (User Datagram Protocol)

UDP is all about speed. It sends data packets out as fast as possible without checking if they arrive. If a packet is lost, it's gone forever.

- **Fast:** No overhead or waiting for acknowledgments.
- **Unordered:** Packets can arrive in any order.
- **Connectionless:** Just fires the data at the destination.
- **Use cases:** Live streaming, online gaming, and VoIP (like WhatsApp calls), where a little lag or jitter is better than a total freeze.

In short: Use TCP when accuracy matters most, and UDP when you need real-time speed! Hope that helps!
